Heng Zhao

PhD Student | Electronic Science and Technology

University of Electronic Science and Technology of China

iezhaoheng@163.com | Chengdu, China

RESEARCH

I study phased-array synthesis and beamforming. My work focuses on sparse and irregular array layouts, wide-angle scanning, and efficient optimization under physical constraints.

EDUCATION

PhD · Electronic Science and Technology | 2023.09 - Present

University of Electronic Science and Technology of China | Integrated master's and doctoral program

Master's studies · Electronic Science and Technology | 2021.09 - 2023.08

University of Electronic Science and Technology of China | Admitted through recommendation

BEng · Electronic Information Engineering | 2017.09 - 2021.06

Zhengzhou University

SELECTED PUBLICATIONS

1. Synthesis of Large-Scale Concentric Ring Arrays for High Sparsity and Wide Scanning via Alternating Optimization

Heng Zhao, Yong-Ling Ban, Lujie Zhou, Yanhui Liu, Jun Hu. IEEE Transactions on Antennas and Propagation, 73(11): 8863-8875, 2025. doi:10.1109/TAP.2025.3591985

2. Scanning Beam Synthesis of Uniform Amplitude Sparse Concentric Ring Arrays Based on Ring Attitude Correction

Heng Zhao, Yongling Ban, Yanhui Liu, Jun Hu. Chinese Journal of Electronics, 34(5): 1507-1521, 2025. doi:10.23919/cje.2024.00.216

3. A Novel Tiling Technique via an Evolutionary Strategy for Large-Scale Irregular Arrays

Heng Zhao, Yong-Ling Ban, Yanhui Liu, Liming Xu, Jun Hu. IEEE Transactions on Antennas and Propagation, 72(4): 3175-3189, 2024. doi:10.1109/TAP.2024.3368195

4. Go-Caterpillar Mutation and Its Optimization Algorithm for Synthesis of Large-Scale Sparse Planar Arrays

Heng Zhao, Yong-Ling Ban, Yanhui Liu, Jun Hu, Zaiping Nie. IEEE Transactions on Antennas and Propagation, 71(3): 2194-2207, 2023. doi:10.1109/TAP.2023.3234658

SELECTED HONORS AND ACTIVITIES

2025 | National Scholarship for Doctoral Students

2024, 2025 | First-class Academic Scholarship, UESTC

2024 | Academic Rising Star Scholarship, UESTC

2025 | Invited presentation | IEEE APCAP | Christchurch, New Zealand

2023 | Academic conference | ACES | Hangzhou, China